Chapter 9

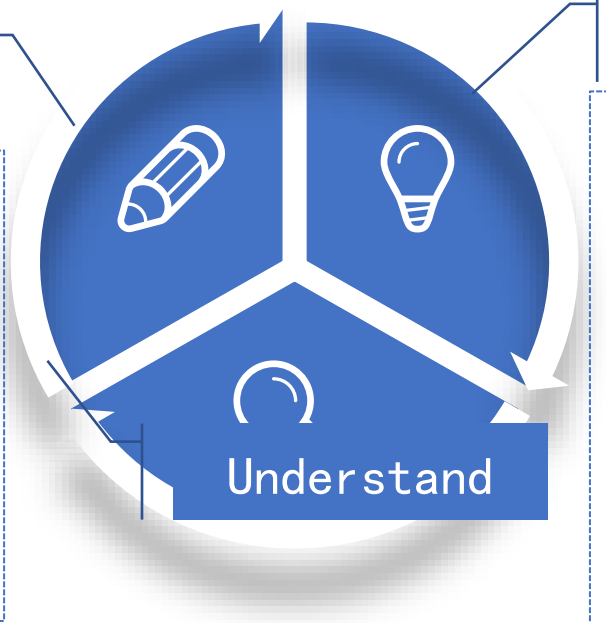
Fluorescence Analysis



C o n t e n t s

- Section 1 Overview
- Section 2 Basic Principles
- Section 3 Instrumentation
- Section 4 Quantitation
- Section 5 New Fluorometric Techniques

Learning Objectives



- Basic principles of fluorometry
- Origin of molecular fluorescence
- Excitation and emission spectra

- Pathways back from the excited state
- Molecular structure and fluorescence
- Factors affecting intensity

- Quantitative methods; spectrofluorimeter; related techniques

Section 1 Overview

Section 1 Overview

- Qualitative and quantitative analysis based on emission spectra is called emission spectrometry利用物质的发射光谱进行定性定量的方法称发射光谱法 （Absorption spectrometry吸收光谱法）

方法名称	辐射能/激发方式	作用物质	检测信号
原子吸收光谱法	紫外、可见光	气态原子外层电子	吸收后的紫外、可见光
紫外-可见光谱法	紫外、可见光	分子外层电子	吸收后的紫外、可见光
红外吸收光谱法	红外光	分子振动	吸收后的红外光
核磁共振波谱法	0.1~900 MHz射频	原子核磁量子等	吸收
原子发射光谱法	火焰、电弧等	气态原子外层电子	紫外、可见光
分子荧光光谱法	紫外、可见光	分子	荧光（紫外、可见光）
磷光光谱法	紫外、可见光	分子	磷光（紫外、可见光）
化学发光法	化学能	分子	可见光
拉曼光谱法	红外光	分子振动	散射后的红外光

Section 1 Overview

Advantages of fluorometry荧光分析法的优点:

- High sensitivity: detection limits of about 10^{-10} g/mL, sometimes 10^{-12} g/mL — three orders of magnitude lower than UV-Vis spectrophotometry
- Several parameters are available — fluorescence lifetime, quantum yield, excitation and emission peak wavelengths — giving good selectivity
- The instrumentation is relatively simple
- 灵敏度高，其检测限一般达10^{-10}g/ml，甚至可达10^{-12}g/ml，比紫外-可见分光光度法低3个数量级以上
- 具有荧光寿命、荧光量子产率、激发峰波长、发射峰波长等多种参数，因而有较好的选择性
- 仪器设备相对较为简单

Section 1 Overview

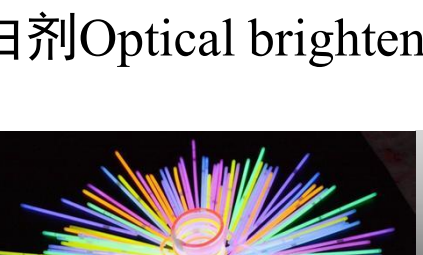
Which ones do you think are true fluorescence:

- 萤火虫Firefly
- 荧光棒Glow stick
- 荧光表（夜光表）Fluorescent watch (luminous watch)
- 荧光灯（日光灯）Fluorescent lamp (daylight lamp)
- 荧光增白剂Optical brightener

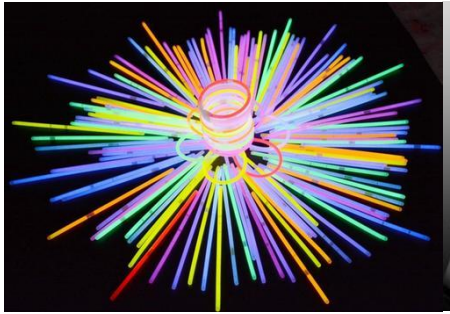
化学发光Chemiluminescence

磷光phosphorescence

荧光fluorescence













Section 2

Basic principles of fluorometry

Section 2 Basic Principles

1 Origin of fluorescence

(1) Electronic energy levels and excitation分子的电子能级与激发过程

- A molecular system has electronic, vibrational, and rotational energy levels. In the ground state, the electrons fill the lowest orbitals in pairs; the two electrons of an orbital have spin quantum numbers +1/2 and −1/2, so the total spin s = 0 and there is no net spin
- 物质的分子体系中存在着电子能级、振动能级和转动能级。在基态，分子中的电子成对地填充在能量最低的各轨道中。给定轨道中的两个电子，自旋量子数分别为1/2和－1/2，其总自旋量子数*s*等于0，即基态没有净自旋

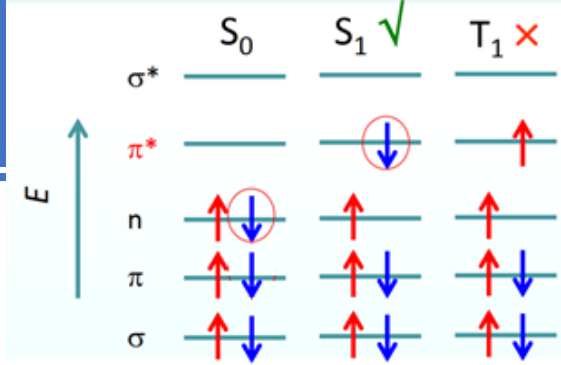
Section 2 Basic Principles

1 Origin of fluorescence

(1) Electronic energy levels and excitation分子的电子能级与激发过程

- The multiplicity of an electronic state is M = 2s + 1. When s = 0, M = 1 and the state is a singlet, denoted S; when s = 1, M = 3 and the state is a triplet, denoted T. The ground state has 2s + 1 = 1
- 电子能级的多重性可用$M = 2s+1$表示，当$s = 0$ 时，分子的多重性$M = 1$，此时分子所处的电子能态称为单重态（singlet state），用符号S表示。当$s = 1$ 时，分子的多重性$M = 3$，此时分子所处的电子能态称为三重态（triplet state），用符号T表示。基态的多重性$2s+1 = 1$

Section 2 Basic Principles



1 Origin of fluorescence

(1) Electronic energy levels and excitation

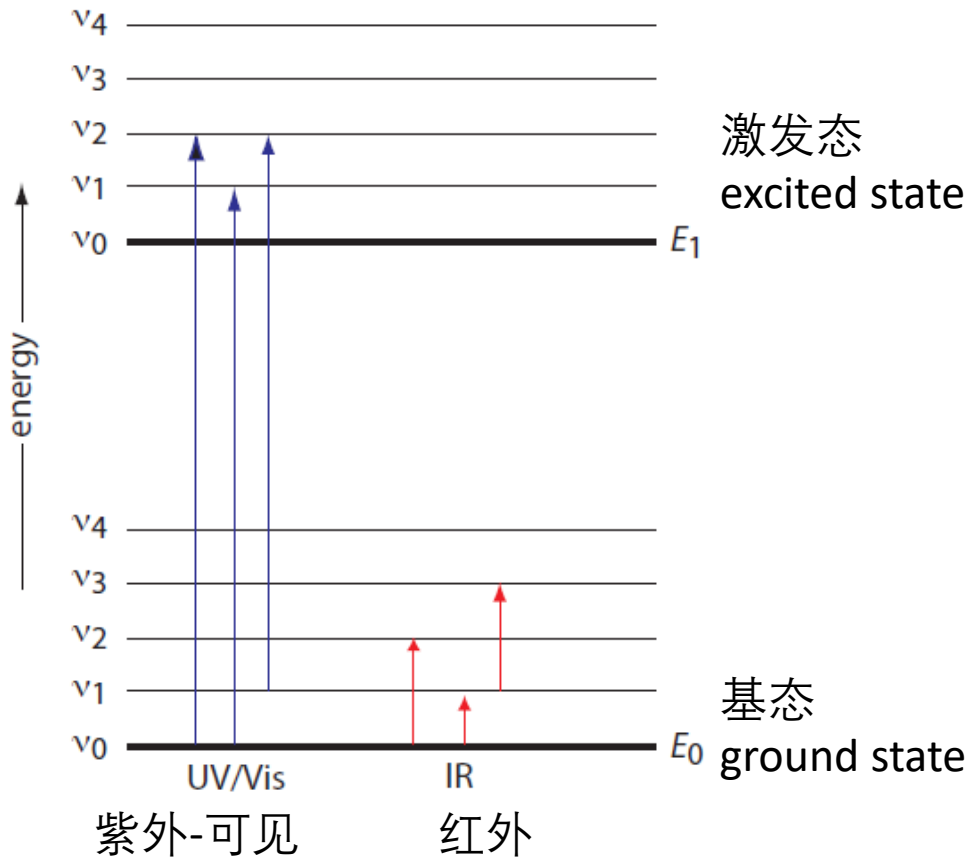
- Singlet state: Every pair of electrons in a molecule has opposite spin directions
- Triplet state: One pair of electrons has the same spin direction
- Allowed transition complying with spectroscopic rules: Transition from the ground state to a higher excited singlet state (e.g., $S_0 \rightarrow S_1$)
- Forbidden transition: Transition from the ground state to an excited triplet state (e.g., $S_0 \rightarrow T_1$)
- 单重态：分子每一对电子自旋方向相反，如基态单重态S_0
- 三重态：有一对自旋方向相同的电子，如第一激发三重态T_1
- 符合光谱规律的跃迁：由基态跃迁至更高的激发单重态（如$S_0 \rightarrow S_1$）
- 禁阻跃迁：由基态跃迁至激发三重态（如$S_0 \rightarrow T_1$）

Section 2 Basic Principles

1 Origin of fluorescence

E 电子能级
electronic
energy level

v 振动能级
vibrational
energy level



Section 2 Basic Principles

1 Origin of fluorescence

(2) Molecular fluorescence and phosphorescence

An excited molecule is unstable and loses energy on returning to the ground state by radiative (light emission) or non-radiative transitions.

电子处于激发态是不稳定状态，返回基态时，通过辐射跃迁（发光）和无辐射跃迁等方式失去能量

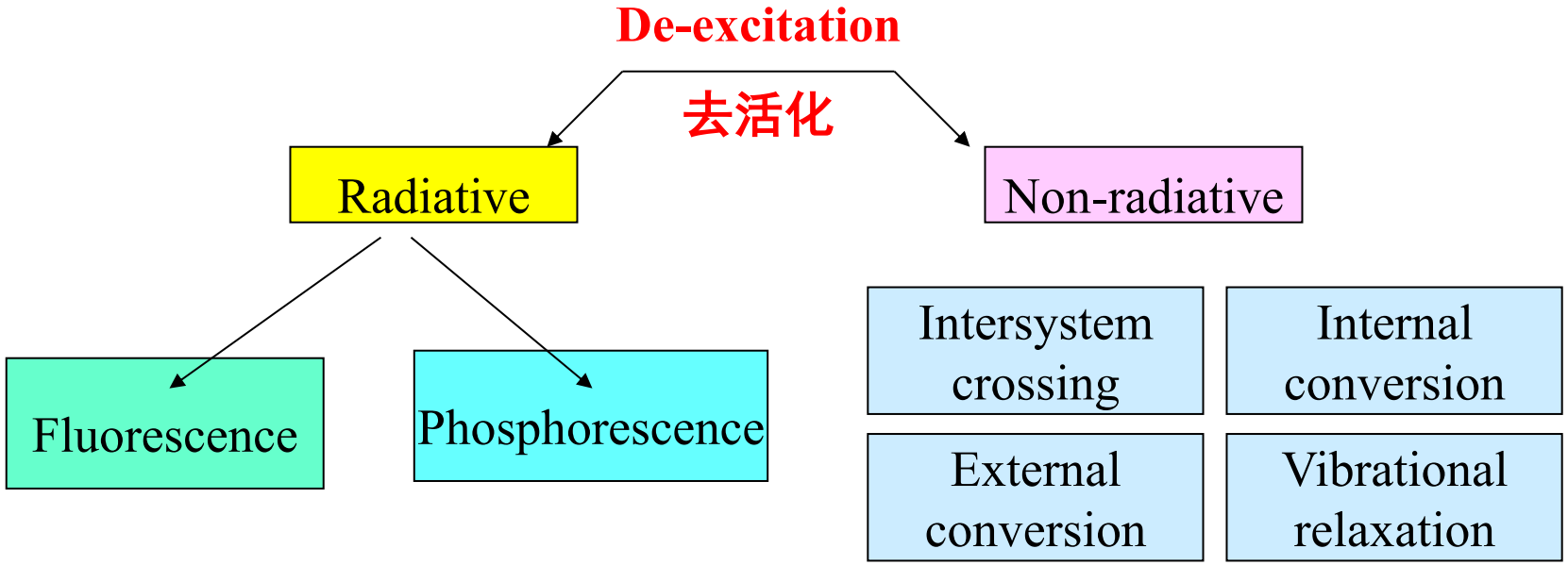
The shorter the stay and the faster the return, the stronger the emission.

激发态停留时间短、返回速度快的途径，发生的概率大，发光强度相对大

Section 2 Basic Principles

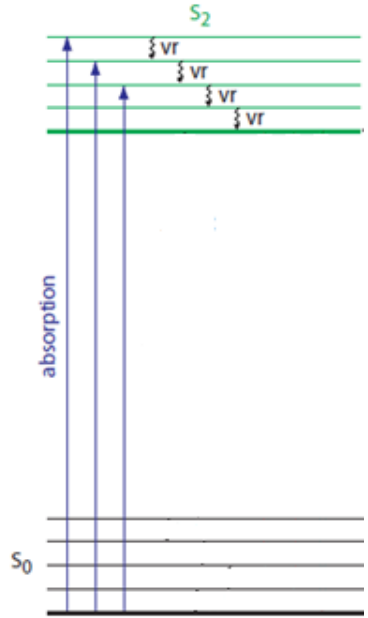
1 Origin of fluorescence

(2) Molecular fluorescence and phosphorescence



Section 2 Basic Principles

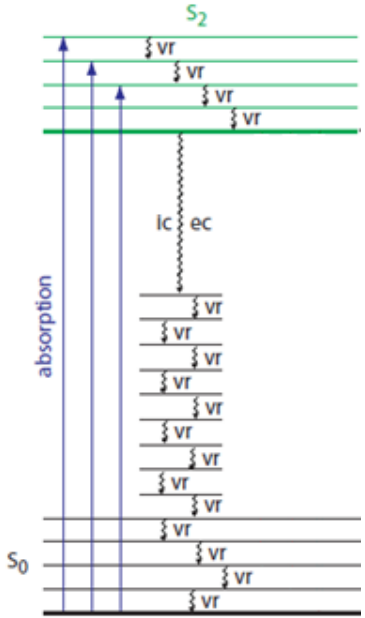
non-radiative transitions



振动弛豫(vr)：vibration relaxation 同一电子能级中，从高振动能层到低振动能层的过程The process of transitioning from a higher vibrational energy level to a lower vibrational energy level within the same electronic energy level

Section 2 Basic Principles

non-radiative transitions

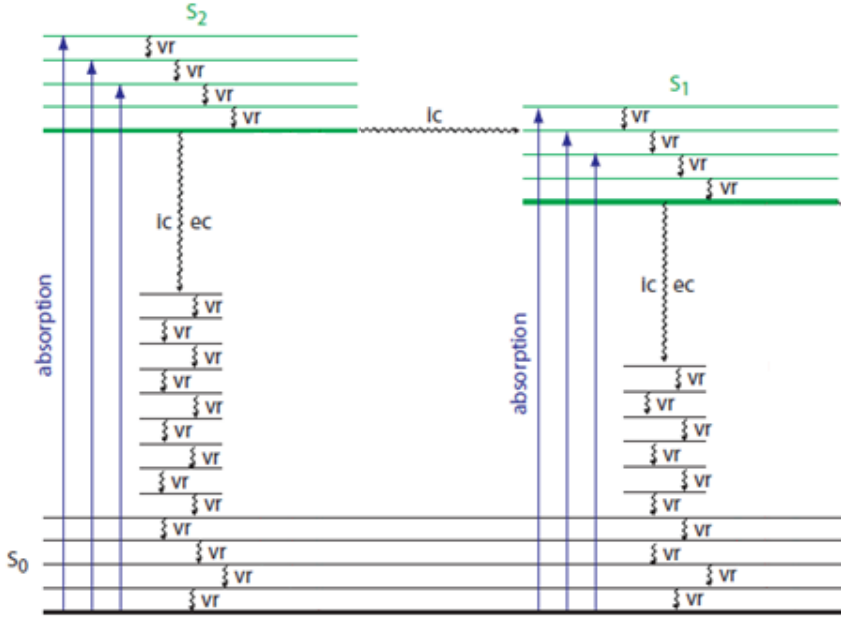


外转换(ec)：external conversion 能量转移至溶剂或溶质分子

Energy transfer to solvent or solute molecules

Section 2 Basic Principles

non-radiative transitions

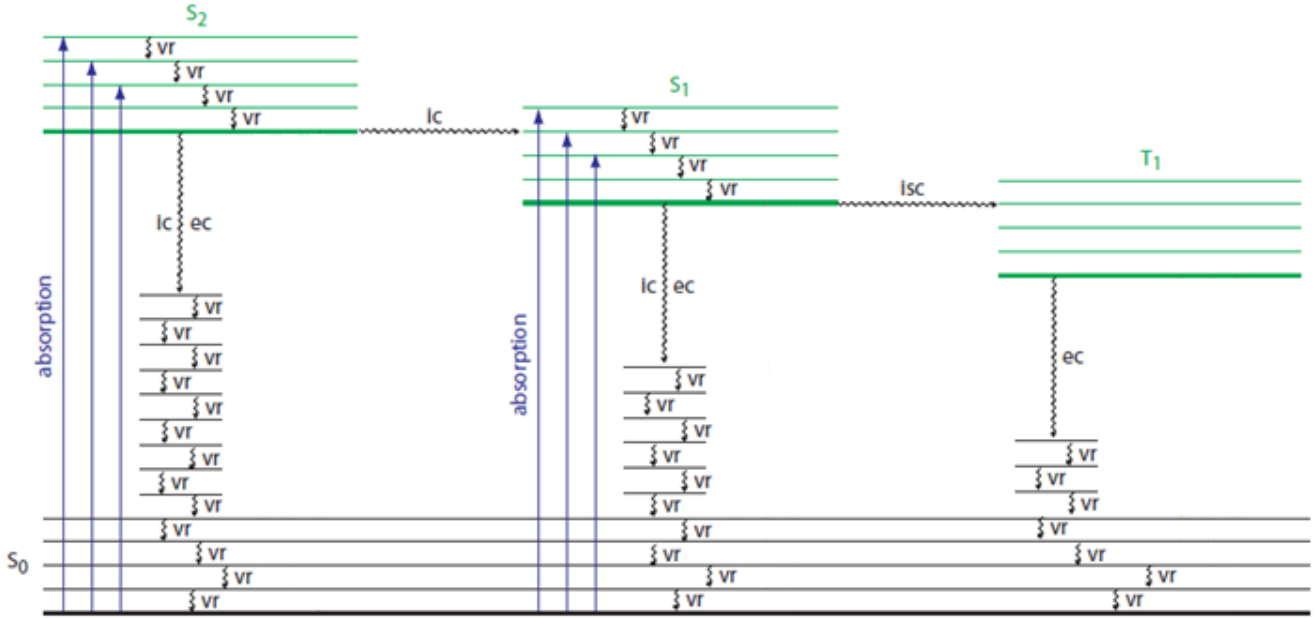


内转换(ic)：internal conversion　　相同的多重态之间的转换，如S_2->S_1

Transitions between identical multiplet states, such as S_2->S_1

Section 2 Basic Principles

non-radiative transitions

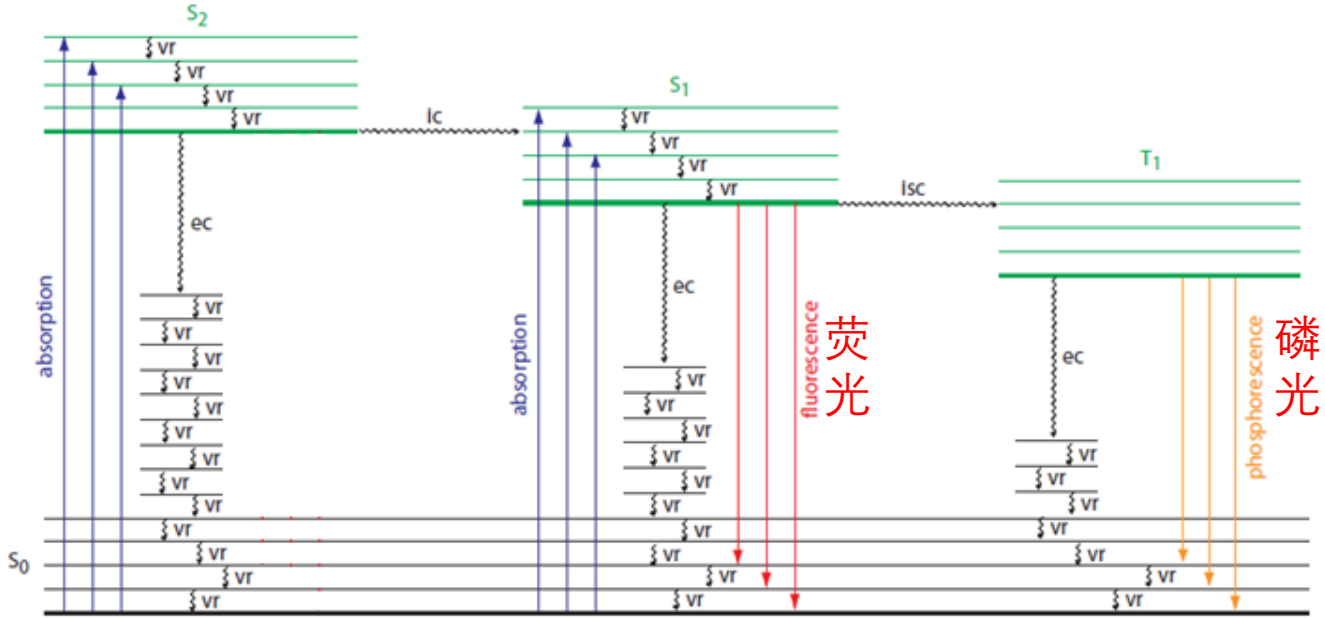


体系间跨越(isc)：intersystem crossing　　不同的多重态之间的转换，如S_1->T_1

Transitions between different multiplet states, such as S_1->T_1

Section 2 Basic Principles

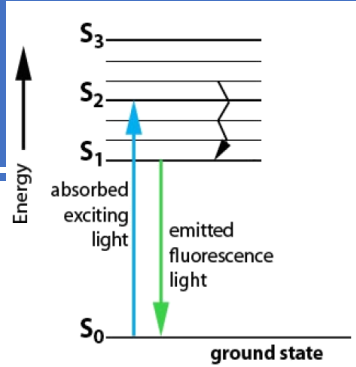
radiative transitions



ec: Energy is transferred to solvent or solute molecules, causing the intensity of fluorescence or phosphorescence to decrease or disappear, which is known as "quenching".

外转换：能量转移至溶剂或溶质分子，荧光或磷光强度减弱或消失，即“猝灭”

Section 2 Basic Principles

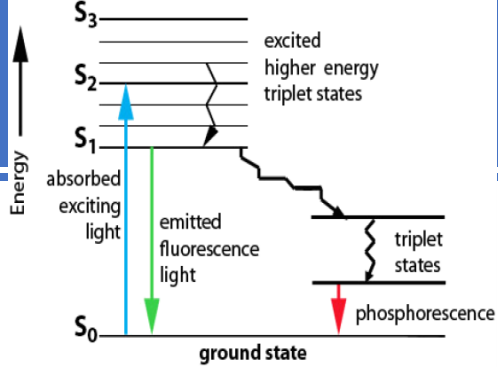


1 Origin of fluorescence

(2) Fluorescence

- Fluorescence refers to the light radiation emitted when electrons return from an excited singlet state to the ground state.电子从激发单重态回到基态所产生的光辐射叫荧光
- Fluorescence is generated rapidly, with a lifetime ranging from 10^{-9} to 10^{-7} seconds. It ceases immediately once the external light source is removed.荧光产生速度快，时间10^{-9}–10^{-7} s，外部光源停止照射，荧光马上熄灭
- Typically, electrons transition from the lowest vibrational level of the first excited singlet state S_1 to various vibrational levels of S_0.通常从第一激发态单重态S_1的最低振动能层回到S_0各振动能层
- The wavelength of fluorescence is longer than that of the excitation light.荧光波长比激发光波长长

Section 2 Basic Principles

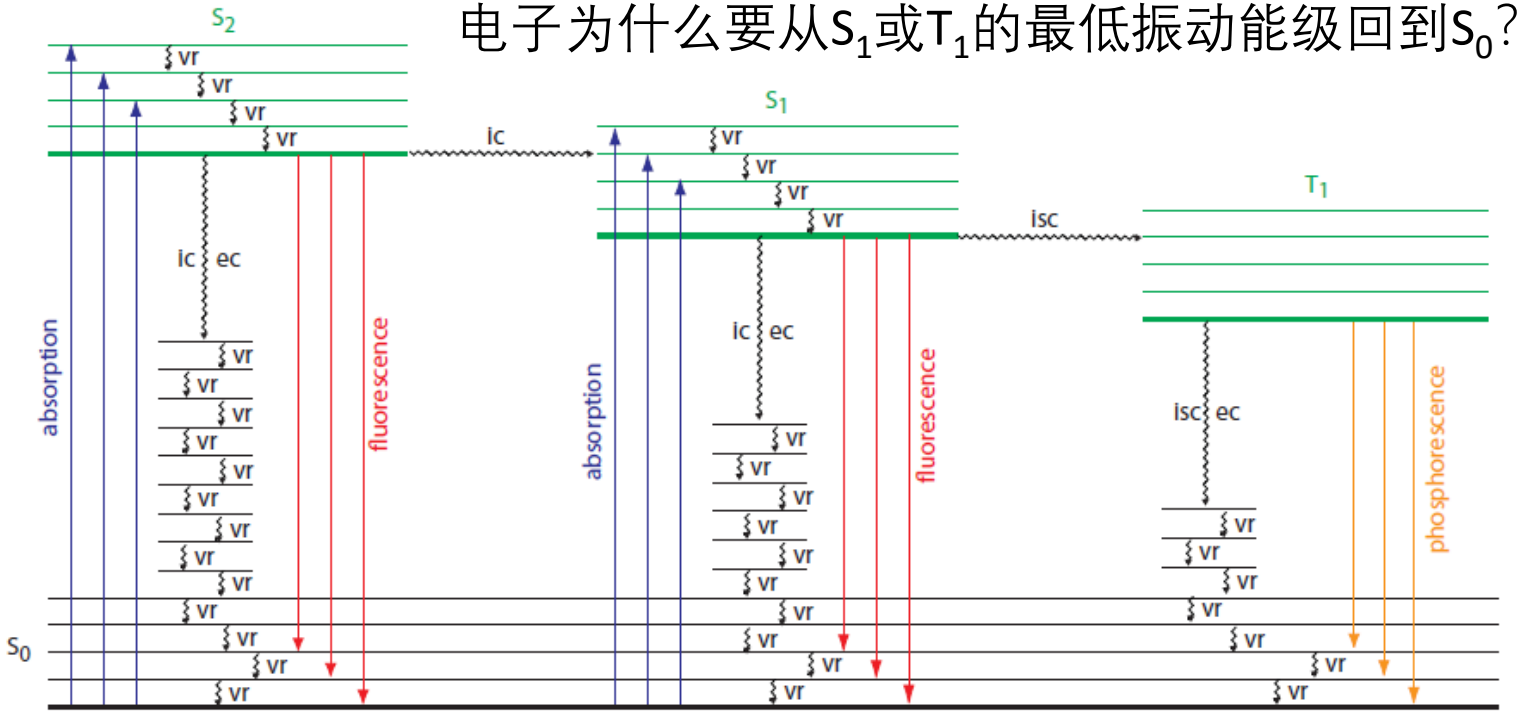


1 Origin of fluorescence

(2) Fluorescence

- The luminescence emitted when electrons return from the excited triplet state to the ground state is called phosphorescence电子从激发三重态回到基态所产生的光辐射叫磷光
- Phosphorescence is generated slowly, with a time range of 10^{-4} to 10^{3} s; after the external light source stops irradiation, phosphorescence can still persist for a period of time磷光产生速度慢，时间10^{-4}–10^{3} s，外部光源停止照射，磷光仍可持续一段时间
- Generally, electrons return from the lowest vibrational energy level of the first excited triplet state T_1 to each vibrational energy level of S_0通常从第一激发态三重态T_1的最低振动能层回到S_0各振动能层
- Phosphorescence has a longer wavelength磷光波长更长

电子为什么要从S_1或T_1的最低振动能级回到S_0?



很快

快

慢

振动弛豫(vr) 10^{-12} s

内转换(ic) 10^{-13}-10^{-11} s

系间跨越(isc) 10^{-10}-10^{-8} s

外转换(ec) 10^{-8}-10^{-5} s

荧光 10^{-9}-10^{-7} s

磷光 10^{-3}-10^{4} s

Section 2 Basic Principles

2 Excitation and fluorescence spectra荧光的激发光谱和荧光光谱

- Excitation spectrum: the relative intensity of fluorescence at a fixed emission wavelength as the excitation wavelength is varied. The emission monochromator is fixed and the excitation monochromator is scanned; the plot of F against excitation wavelength λ_{ex} is the excitation spectrum, closely resembling the absorption spectrum
- 激发光谱（excitation spectrum）：不同激发波长的辐射引起物质发射某一波长荧光的相对强度。固定发射单色器在某一波长，通过激发单色器扫描，以不同波长的入射光激发荧光物质，记录荧光强度（F）对激发波长（λ_{ex}）的关系曲线，即激发光谱，其形状与其吸收光谱极为相似

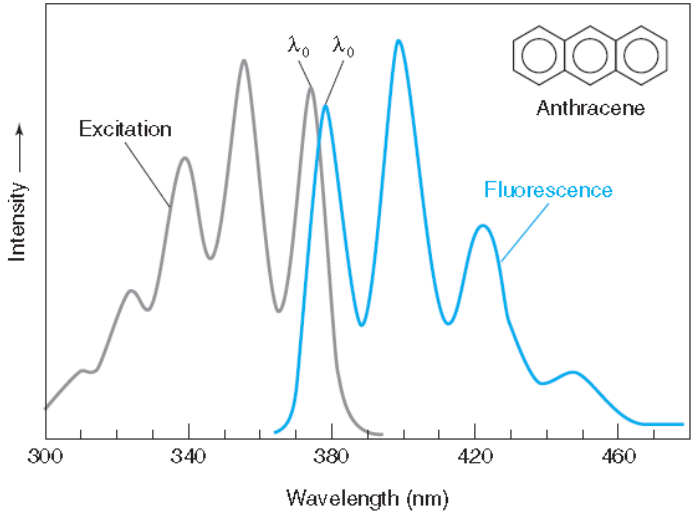
2 Excitation and fluorescence spectra荧光的激发光谱和荧光光谱

- Emission spectrum: the relative intensities of the wavelength components of the emitted fluorescence, also called the fluorescence spectrum. The excitation wavelength and intensity are held constant while the emission monochromator is scanned, giving F against λ_{em}
- 发射光谱（emission spectrum）：所发射的荧光中各种波长组分的相对强度，或称荧光光谱（fluorescence spectrum）。激发光的波长和强度保持不变，通过发射单色器扫描，记录荧光强度（F）对发射波长（λ_{em}）的关系曲线

Section 2 Basic Principles

2 Excitation and fluorescence spectra荧光的激发光谱和荧光光谱

- Excitation and fluorescence spectra reflect structural features, so they identify fluorescent substances and indicate the wavelengths to use
- 激发光谱和荧光光谱反映了物质的结构特征，因此，可用于鉴别荧光物质，而且是选择测定波长的依据



Section 2 Basic Principles

2 Excitation and fluorescence spectra

Solution fluorescence spectra generally show:

1. Stokes shift斯托克斯位移: the emission wavelength is always longer than the excitation wavelength.是荧光发射波长总是大于激发光波长的现象。





Figure 1.7. Sir George Gabriel Stokes, 1819–1903, Lucasian Professor at Cambridge. Reproduced courtesy of the Library and Information Centre, Royal Society of Chemistry.

2 Excitation and fluorescence spectra

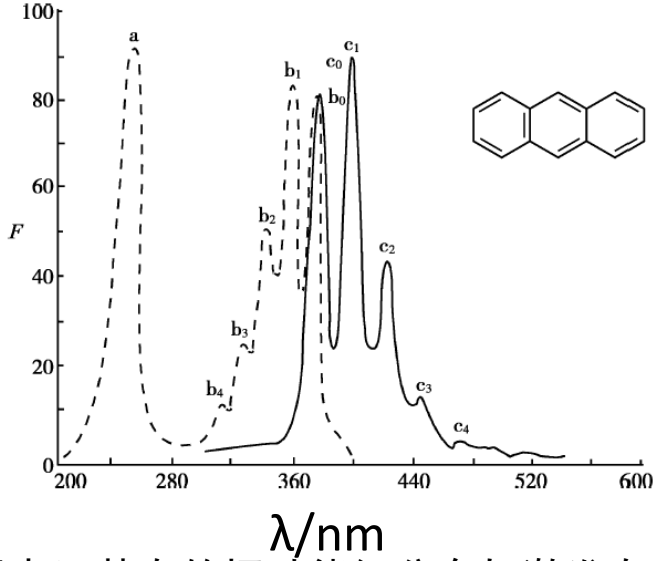
Solution fluorescence spectra generally show:

2. The shape of the fluorescence spectrum is independent of the excitation wavelength荧光光谱的形状与激发波长无关： Even if the molecule is excited above S_1^*, fast internal conversion and vibrational relaxation bring it to the lowest vibrational level of S_1^*即使分子被激发到高于S_1^*的电子激发态的各个振动能级，由于内转换和振动弛豫的速度很快，都会下降至S_1^*的最低振动能级

Section 2 Basic Principles

2 Excitation and fluorescence spectra

3. Mirror-image relation荧光光谱与



由于电子基态的振动能级分布与激发态相似，故激发光谱与荧光光谱的形状相似，而且两者之间存在“镜像对称”关系

各小峰间波长递减值Δλ与振动能级差Δ*E*有关
各小峰的高度与跃迁概率有关

Δλ between sub-peaks ↔ vibrational ΔE
peak height ↔ transition probability

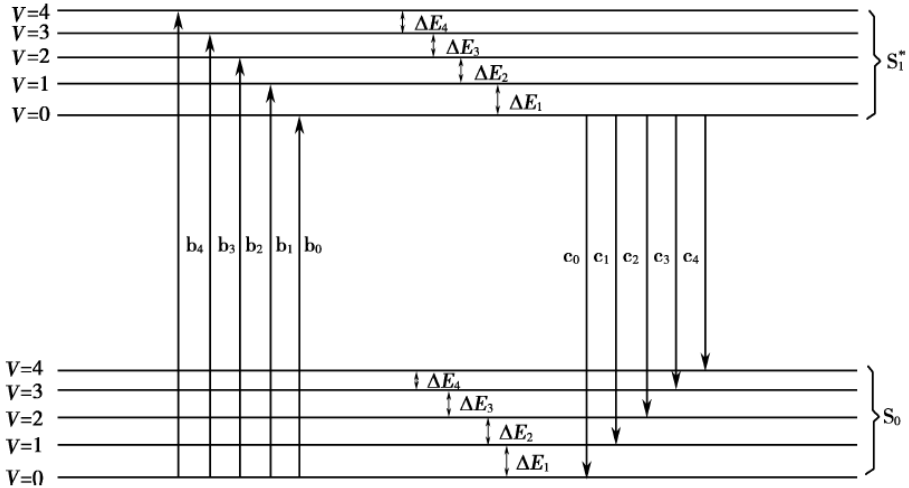


图 12-5 蒽的能级跃迁

3 Fluorescence and molecular structure荧光与分子结构的关系

Fluorescence lifetime: after the excitation source is removed, the time for the intensity to fall to 1/e of its maximum, usually written τ_f.

荧光寿命（fluorescence life time）是当除去激发光源后，分子的荧光强度降低到最大荧光强度的1/e所需的时间，常用τ_f表示。

The decay follows an exponential law:

$$F_t = F_0 e^{-Kt} \quad (1)$$

式中F_0和F_t分别是在激发时$t = 0$和激发后时间t时的荧光强度，K是衰减常数

3 Fluorescence and molecular structure

(1) Fluorescence lifetime and efficiency

If at $t = \tau_f$ the intensity has fallen to 1/e of F_0, i.e. $F_t = (1/e)F_0$, then

$$\frac{F_0}{e} = F_0 e^{-K\tau_f}$$

$$1/e = e^{-K\tau_f},\quad K\tau_f = 1,\quad K = \frac{1}{\tau_f}$$

Equation (1) becomes $F_t/F_0 = e^{-Kt}$, giving $\ln\frac{F_0}{F_t} = \frac{t}{\tau_f}$

A plot of $\ln(F_0/F_t)$ against t is a straight line whose slope gives $1/\tau_f$. Differences in lifetime allow mixtures of fluorescent substances to be analysed 利用分子荧光寿命的差别，可以进行荧光物质混合物的分析

3 Fluorescence and molecular structure

(1) Fluorescence lifetime and efficiency

Fluorescence efficiency, also called the fluorescence quantum yield: the ratio of photons emitted to photons absorbed, usually written φ_f:

荧光效率（fluorescence efficiency）又称荧光量子产率（fluorescence quantum yield），是指激发态分子发射荧光的光子数与基态分子吸收激发光的光子数之比，常用φ_f表示：

$$\varphi_f = \frac{\text{发射荧光的光子数}}{\text{吸收激发光的光子数}}$$

3 Fluorescence and molecular structure

(1) Fluorescence lifetime and efficiency

Quantum yields lie between 0 and 1: sodium fluorescein in water $\varphi_f = 0.92$; fluorescein in water$\varphi_f = 0.65$; anthracene in ethanol $\varphi_f = 0.30$; phenanthrene in ethanol $\varphi_f = 0.10$.

一般物质的荧光效率在0～1之间。例如荧光素钠在水中$\varphi_f = 0.92$；荧光素在水中$\varphi_f = 0.65$；蒽在乙醇中$\varphi_f = 0.30$；菲在乙醇中$\varphi_f = 0.10$。荧光效率低的物质所吸收的能量都以无辐射跃迁形式释放，所以没有荧光发射

A substance with a low quantum yield loses all absorbed energy non-radiatively and does not fluoresce荧光效率低的物质所吸收的能量都以无辐射跃迁形式释放，所以没有荧光发射

3 Fluorescence and molecular structure

(2) Structure of organic compounds and fluorescence

Requirements for fluorescence

1. strong UV-Vis absorption — conjugated $\pi \rightarrow \pi^*$ (K band)

1.有强的紫外-可见吸收　共轭$\pi \rightarrow \pi^*$（K带）

2.有一定的荧光效率　刚性平面结构

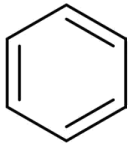
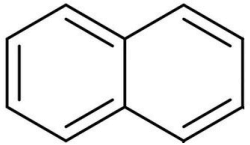
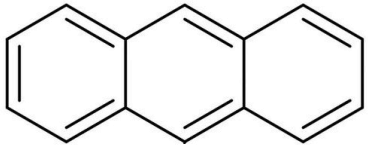
2. a reasonable quantum yield — a rigid, planar structure

Substituents on the conjugated system also strongly affect the spectrum and the intensity共轭体系上的取代基对荧光光谱和荧光强度也有很大影响

3 Fluorescence and molecular structure

1. Long conjugated structure长共轭结构

Aromatic or heterocyclic rings with long π→π* conjugation fluoresce strongly; more conjugation gives stronger, longer-wavelength emission.绝大多数能产生荧光的物质都含有芳香环或杂环（长共轭的π→π*跃迁）。π电子共轭程度↑，荧光强度（荧光效率）↑，而荧光波长↑

	benzene 苯	萘 naphthalene	蒽 anthracene
			
λ_{ex}	205nm	286nm	356nm
λ_{em}	278nm	321nm	404nm
ϕ_f	0.11	0.29	0.36

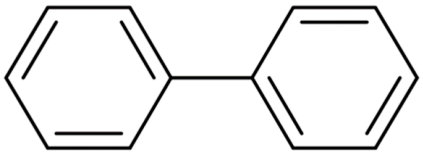
3 Fluorescence and molecular structure

2. molecular rigidity分子的刚性

In the same long-conjugated molecules, the greater the rigidity of the molecule, the higher the fluorescence quantum yield and the longer the fluorescence wavelength.

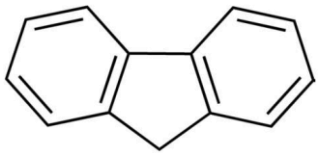
在同样的长共轭分子中，分子的刚性↑，荧光效率↑，荧光波长↑

联苯



ϕ_f 0.2

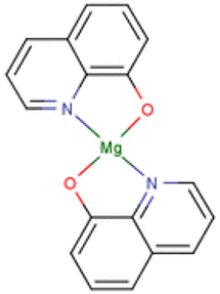
芴

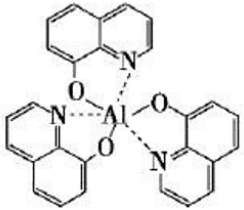


1.0



8-羟基喹啉





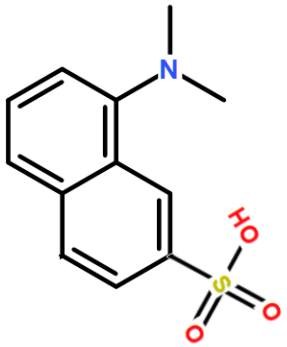
A weakly fluorescent or non-fluorescent substance may fluoresce strongly after forming a rigid, coplanar complex with a metal ion, 8-hydroxyquinoline with Mg^{2+} or Al^{3+}

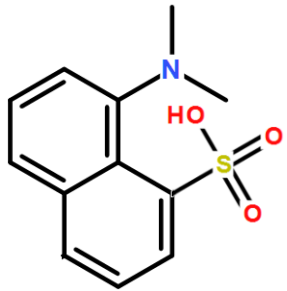
3 Fluorescence and molecular structure

2. molecular rigidity分子的刚性

If the original structure has good coplanarity, but steric hindrance reduces the molecular coplanarity, the fluorescence will weaken.

如果原来结构中共平面性较好，但由于位阻效应使分子共平面性下降，则荧光减弱





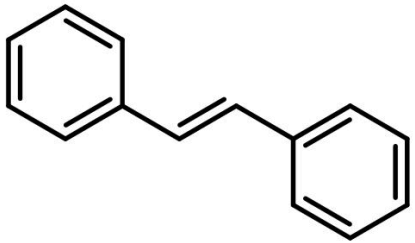
1-二甲氨基萘-7-磺酸	1-二甲氨基萘-8-磺酸
ϕ_f 0.75	0.03

3 Fluorescence and molecular structure

2. molecular rigidity分子的刚性

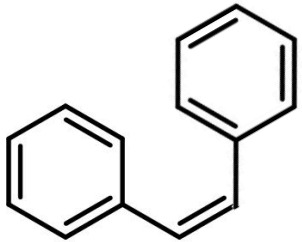
For cis-trans isomers, the two groups of the cis molecule are on the same side; steric hindrance prevents the molecule from being coplanar, resulting in no fluorescence.

对于顺反异构体，顺式分子的两个基团在同一侧，由于位阻效应使分子不能共平面而没有荧光



反-1,2-二苯乙烯

√



顺-1,2-二苯乙烯

×√

3 Fluorescence and molecular structure

(2) Structure of organic compounds and fluorescence

3. Substituents取代基

- Electron-donating groups: increase conjugation给电子基团：增加分子的电子共轭程度— $\varphi_f \uparrow$, $\lambda_{em} \uparrow$
  —NH_2, —OH, —OCH_3, —NHR, —CN, etc.
- Electron-withdrawing groups: decrease π conjugation给电子基团：增加分子的电子共轭程度— $\varphi_f \downarrow$
  —COOH, —NO_2, —C=O, —NO, —SH, —$NHCOCH_3$, —X, etc.
- Groups with little effect on the π system对π电子共轭体系作用较小的基团— φ_f unchanged
  —R, —SO_3H, —NH_3^+, etc.

3 Fluorescence and molecular structure

(3) Fluorogenic reagents荧光试剂

To improve sensitivity and selectivity, weakly fluorescent substances are converted into strongly fluorescent products with fluorogenic reagents, extending the scope of fluorometry

为了提高测定的灵敏度和选择性，常使弱荧光物质与某些荧光试剂作用，以得到强荧光性产物，扩大荧光分析法的应用范围

Section 2 Basic Principles

3 Fluorescence and molecular structure

(3) Fluorogenic reagents荧光试剂

1. Fluorescamine荧光胺

forms strongly fluorescent derivatives with aliphatic and aromatic primary amines与脂肪族和芳香族伯胺形成高强度的荧光衍生物

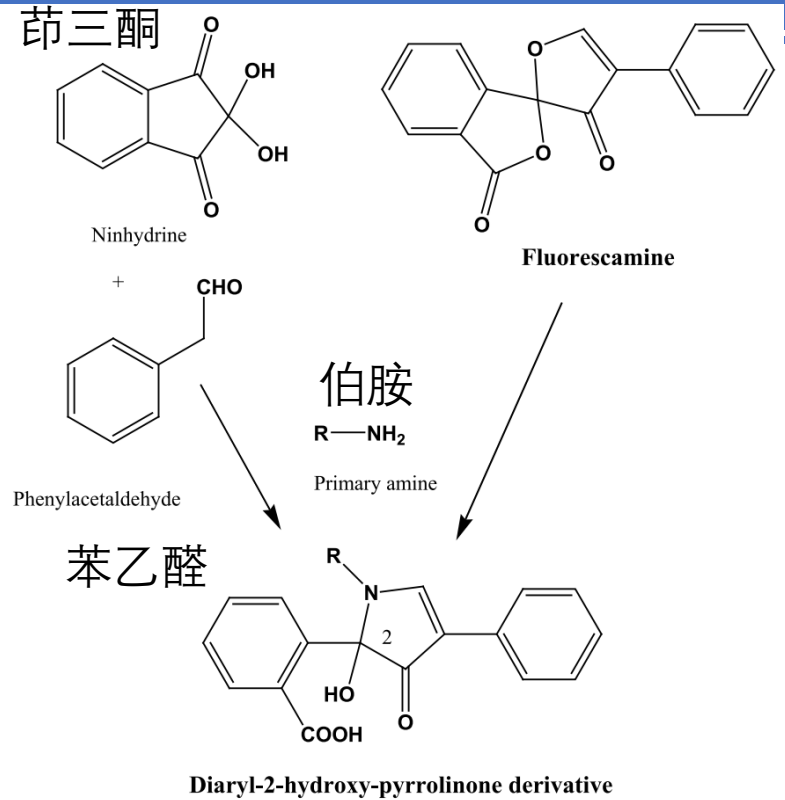


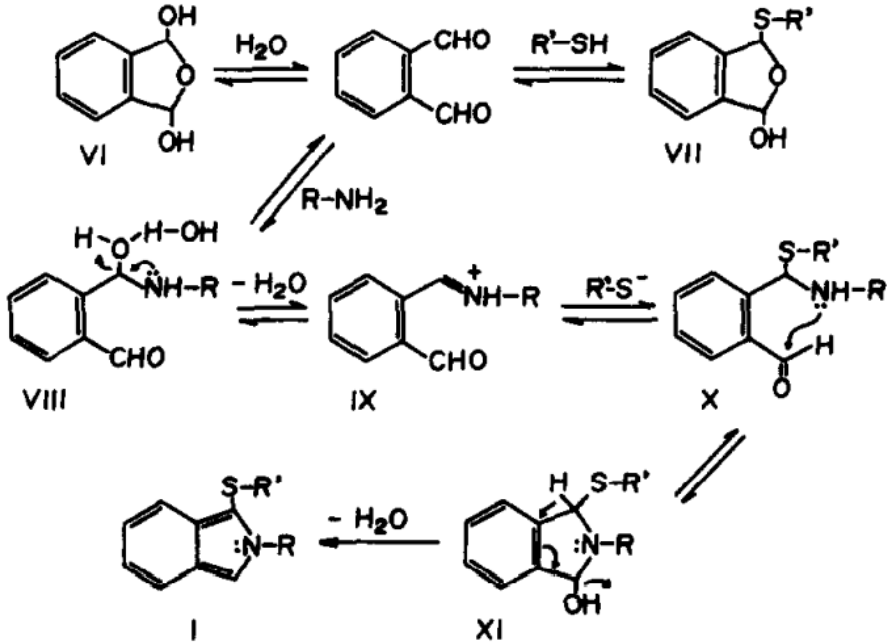
Fig. 1. Condensation of primary amines with ninhydrin and aldehydes, and the nucleophilic substitution reaction between primary amine and Fluorescamine (FSA).

3 Fluorescence and molecular structure

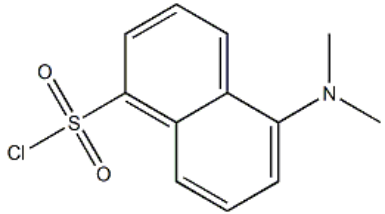
(3) Fluorogenic reagents荧光试剂

2. o-Phthalaldehyde (OPA)

gives sensitive fluorescent products with primary amines, especially α-amino acids other than cysteine, proline and hydroxyproline与伯胺类，特别是半胱氨酸、脯氨酸及羟脯氨酸外的α-氨基酸生成灵敏的荧光产物



3 Fluorescence and molecular structure



(3) Fluorogenic reagents

3. Dansyl chloride (5-dimethylamino-1-naphthalenesulfonyl chloride)

1-二甲氨基-5-氯化磺酰萘（Dansyl-Cl，丹酰氯）

reacts with primary and secondary amines and with phenolic alkaloids to give fluorescent products与伯、仲胺及酚基的生物碱反应生成荧光产物

4. Reagents for inorganic ions测定无机离子的荧光试剂

4 External factors affecting intensity影响荧光强度的外部因素

1. Temperature

temperature ↑, φ_f ↓

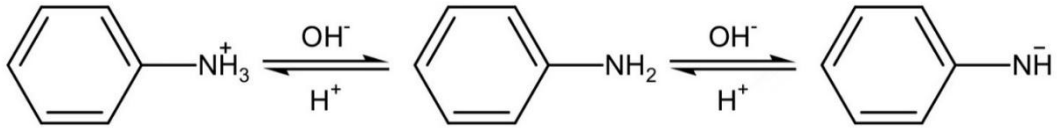
2. Solvent

solvent polarity ↑, φ_f ↑, λ_{em} ↑; solvent viscosity ↓, φ_f ↓



3. Acidity

many fluorophores are weak acids or bases



pH < 2 pH 7-12 pH > 13

4 External factors affecting intensity

4. Quenchers荧光熄灭剂

- Fluorescence quenching is the decrease in intensity caused by interaction of the fluorophore with the solvent or other solute molecules; the species responsible is a quencher.荧光熄灭又称荧光猝灭，是指荧光物质分子与溶剂分子或其他溶质分子相互作用引起荧光强度降低的现象。引起荧光熄灭的物质称为荧光熄灭剂
- At high concentration (above 1 g/L) self-quenching also occurs浓度较大（超过1g/L）时，还会发生自熄灭现象

4 External factors affecting intensity

4. Quenchers

Fluorescence quenching method: after adding a quencher, the decrease in intensity is linear in the quencher concentration, so the quencher can be determined. For example, the quenching of the borate-benzoin complex by oxygen allows trace oxygen to be measured

荧光熄灭法（fluorescence quenching method）：一个荧光物质在加入某种熄灭剂后，荧光强度的减弱和荧光熄灭剂的浓度呈线性关系，利用这一性质测定荧光熄灭剂的含量。如利用氧分子对硼酸根-二苯乙醇酮配合物的荧光熄灭效应，可进行微量氧的测定

4 External factors affecting intensity

5. Scattering light散射光

- When a parallel monochromatic beam strikes a liquid sample, most of the light is transmitted and a small part is scattered at various angles after collisions with molecules一束平行单色光照射在液体样品上时，大部分光线透过溶液，小部分由于光子和物质分子相碰撞，使光子的运动方向发生改变而向不同角度散射

Section 2 Basic Principles

4 External factors affecting intensity

5. Scattering light散射光

- Rayleigh scattering: elastic collisions with molecules, no energy exchange, only a change of direction; the wavelength equals that of the incident light
- 瑞利散射光（Rayleigh scattering light）：光子和物质分子发生弹性碰撞，不发生能量的交换，仅仅是光子运动方向发生改变。波长与入射光波长相同

拉曼光谱入门之一: 光的散射_哔哩哔哩_bilibili

Section 2 Basic Principles

4 External factors affecting intensity

5. Scattering light散射光

- Raman scattering: inelastic collisions in which the direction changes and energy is exchanged — the photon loses energy to, or gains energy from, the molecule, so the scattered light is slightly longer or shorter in wavelength
- Raman light of longer wavelength than the incident light interferes more strongly
- 拉曼散射光（Raman scattering light）：光子和物质分子发生非弹性碰撞时，在光子运动方向发生改变的同时，光子与物质分子发生能量的交换，光子把部分能量转移给物质分子或从物质分子获得部分能量，而发射出比入射光稍长或稍短的光
- 波长比入射光波长更长的拉曼光，对荧光测定的干扰更大

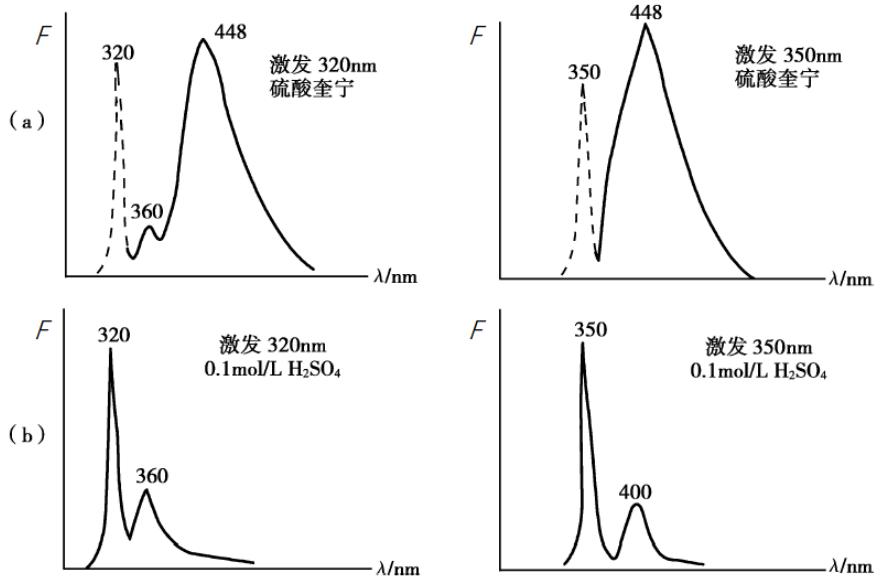
拉曼光谱入门系列之九：拉曼光谱的激光_哔哩哔哩_bilibili

Section 2 Basic Principles

4 External factors affecting intensity

A suitable excitation wavelength removes the Raman interference

选择适当的激发波长可消除拉曼光的干扰



Fluorescence of quinine sulfate at different excitation wavelengths (upper) and scattering spectra of the solvent (lower)

硫酸奎宁在不同激发波长下的荧光（上）与溶剂的散射光谱（下）

4 External factors affecting intensity

Raman wavelengths of common solvents /nm

Solvent	Excitation /nm				
	248	**313**	**365**	**405**	**436**
water	271	350	416	469	511
ethanol	267	344	409	459	500
cyclohexane	267	344	408	458	499
CCl_4	—	320	375	418	450
$CHCl_3$	—	346	410	461	502

Section 3 Instrumentation

Section 3 Instrumentation

1 Components of the instrument

Instruments for measuring fluorescence:

- **Filter fluorimeter滤光片荧光计**
  - the excitation filter passes the excitation light, the emission filter passes only the sample fluorescence
  - spectra cannot be recorded, but quantitation is possible
  - 激发滤光片让激发光通过；发射滤光片只允许试样的荧光通过
  - 不能测定光谱，但可用于定量分析

Section 3 Instrumentation

1 Components of the instrument

Instruments for measuring fluorescence:

- **Filter-monochromator fluorimeter**滤光片-单色器荧光计
  - the emission filter is replaced by a grating
  - the excitation spectrum cannot be recorded, but the fluorescence spectrum can
  - 将发射滤光片用光栅代替
  - 不能测定激发光谱，但可测定荧光光谱

Section 3 Instrumentation

1 Components of the instrument

Instruments for measuring fluorescence:

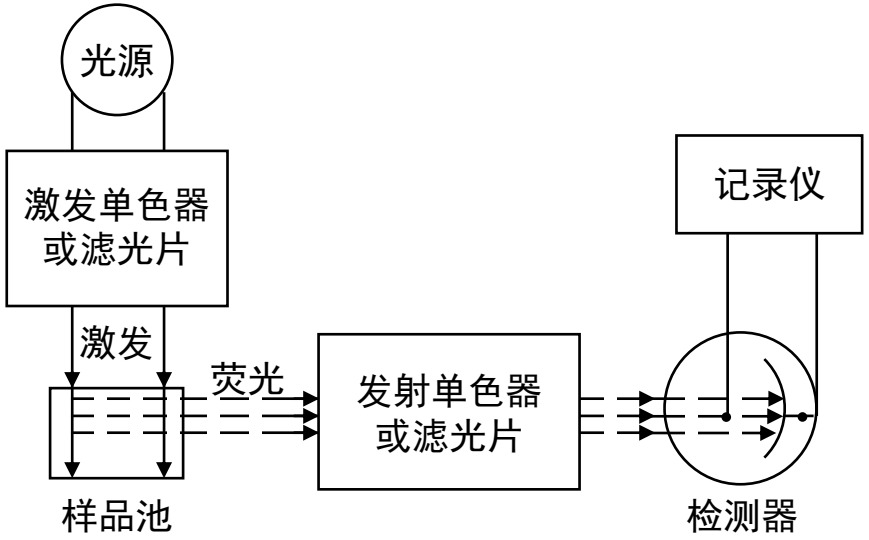
- **Spectrofluorimeter**
  - Both filters are replaced with gratings
  - measures the intensity at one wavelength and records both spectra
  - 两个滤光片都用光栅取代
  - 既可测量某一波长处的荧光强度，还可绘制激发光谱和荧光光谱

Section 3 Instrumentation

1 Components of the instrument

Main components:

- excitation source: xenon lamp
  激发光源：氙灯, 250-700 nm
- monochromators单色器: filters or gratings滤光片，光栅
- sample cell样品池: quartz石英
- detection: photomultiplier tube
  检测系统：光电倍增管



Block diagram of a spectrofluorimeter

2 Calibration of the instrument荧光分光光度计的校正（不要求）

(1) sensitivity calibration

(2) wavelength calibration

(3) correction of excitation and fluorescence spectra

Section 4 Quantitation

Section 4 Quantitation

1 Intensity and concentration溶液荧光强度与物质浓度的关系

The fluorescence intensity depends on how much light the fluorophore absorbs and on its quantum yield.溶液的荧光强度与该溶液中荧光物质吸收光能的程度以及荧光效率有关

- 设溶液中荧光物质浓度为c，液层厚度为l
- The fluorescence intensity is proportional to the intensity of light absorbed by the fluorescent substance.荧光强度正比于被荧光物质吸收的光强度
  - $F \propto K'(I_0-I)$，K'为常数，取决于荧光效率
  - $I=I_0 10^{-Ecl}$，当$Ecl \leq 0.05$时，$F=Kc$，K为常数
- At low concentrations, F has a linear relationship with concentration c低浓度时F与浓度c成线性关系

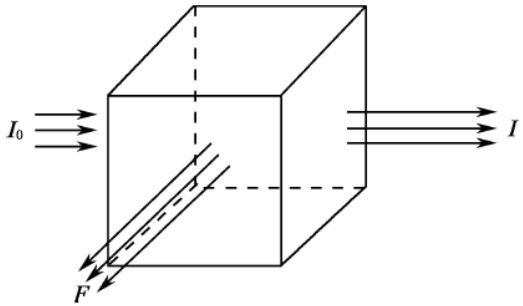


图 12-7 溶液的荧光测定

Section 4 Quantitation

1 Intensity and concentration溶液荧光强度与物质浓度的关系

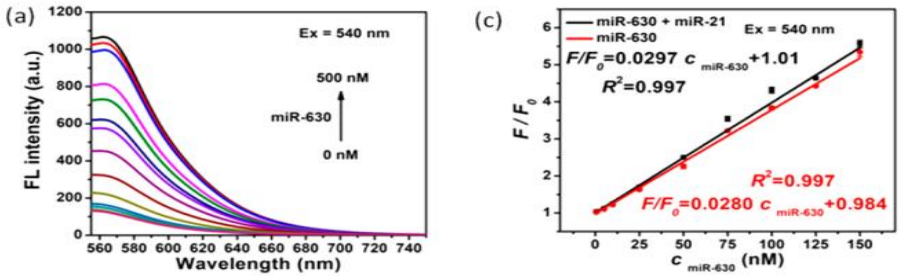
	荧光分析	紫外-可见吸收
物质浓度关系	荧光强度F=Kc $K=2.3K'I_0Ecl$	吸光度A=Ecl $A=-lg(I/I_0)$
荧光强度/吸光度影响因素	荧光效率，吸光系数，浓度，光程	吸光系数，浓度，光程
测定值	弱背景下的荧光强度（发射光强度）	透过光强与入射光强的比制
灵敏度	10^{-10}~10^{-12}g/mL	10^{-4}~10^{-7}g/mL

Section 4 Quantitation

2 Direct methods

1. Calibration curve method工作曲线法

- Plot fluorescence intensity against the concentration of the standards
- 以荧光强度为纵坐标，对照品溶液的浓度为横坐标绘制工作曲线
- After zeroing, measure the blank and then the standards:
- 仪器调零之后，先测定空白溶液的荧光强度，然后测定对照品溶液的荧光强度
- $F = F_{standard} - F_{blank}$



Anal. Chem. 2020, 92, 11565−11572

2 Direct methods

2. Proportional method比例法

- If the calibration curve passes through the origin, a single standard within the linear range may be used
- 如果荧光分析的工作曲线通过原点，就可选择其线性范围，用比例法进行测定
- If the blank cannot be set to zero, subtract the blank intensity F_0 from F_s and F_x before calculating
- 在空白溶液的荧光强度调不到0时，必须从F_s及F_x值中扣除空白溶液的荧光强度（F_0），然后进行计算

$$\frac{F_s - F_0}{F_x - F_0} = \frac{c_s}{c_x} \qquad c_x = \frac{F_x - F_0}{F_s - F_0} \times c_s$$

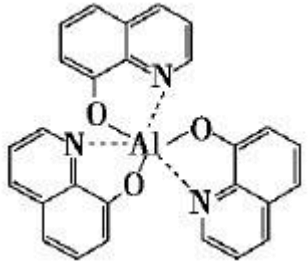
2 Direct methods

2. Proportional method比例法

- 氯化钠中铝盐的测定（中国药典）
- 铝被人体吸收后取代镁离子与生物配体结合，抑制酶活性、使神经纤维变性、干扰神经系统功能。
- 由于铝元素主要由肾脏排泄，维持性血液透析患者肾脏排泄功能障碍，导致慢性中毒的血清铝蓄积，高出正常水平14倍。
- 因此，需要对供制备血液透析液、血液过滤液或腹膜透析液用的氯化钠中的铝盐进行检查。

Section 4 Quantitation

2 Direct methods



2. Proportional method比例法

- 氯化钠中铝盐的测定（中国药典）
- 原理：铝离子在合适的pH下可以与8-羟基喹啉形成稳定的荧光络合物
- 1. 配置被测样品（氯化钠）溶液
- 取氯化钠20.0g，加水100ml溶解，再加入醋酸-醋酸铵缓冲液（pH6.0）10ml。将上述溶液移至分液漏斗中，加入0.5%的8-羟基喹啉三氯甲烷溶液提取三次（20ml、20ml、10ml），合并提取液，置50ml量瓶中，加三氯甲烷至刻度，摇匀。

Section 4 Quantitation

2 Direct methods

2. Proportional method比例法

- 氯化钠中铝盐的测定（中国药典）
- 2. 配置对照溶液
- 精密量取铝单元素标准溶液适量，用2%硝酸溶液定量稀释制成每1ml中含铝（Al）2μg的标准溶液。取上述铝标准溶液2.0ml，加水98ml和醋酸-醋酸铵缓冲液（pH 6.0）10ml。将上述溶液移至分液漏斗中，加入0.5%的8-羟基喹啉三氯甲烷溶液提取三次（20ml、20ml、10ml），合并提取液，置50ml量瓶中，加三氯甲烷至刻度，摇匀。

2 Direct methods

2. Proportional method比例法

- 氯化钠中铝盐的测定（中国药典）
- 3. 配置空白溶液
- 量取醋酸-醋酸铵缓冲液（pH 6. 0）10ml, 加水100ml。将上述溶液移至分液漏斗中，加入0.5%的8-羟基喹啉三氯甲烷溶液提取三次（20ml、20ml、10ml），合并提取液，置50ml量瓶中，加三氯甲烷至刻度，摇匀。

2 Direct methods

2. Proportional method比例法

- 氯化钠中铝盐的测定（中国药典）
- 4. 测定荧光强度
- 取上述三种溶液，使用荧光分光光度计在激发波长392nm下、测定发射波长518nm处的荧光强度。
- 5. 判定检测结果
- 被测样品溶液的荧光强度应不大于对照溶液的荧光强度，即铝含量不大于0.00002%。

3 Indirect methods间接测定法（不要求）

Substances that do not fluoresce, or whose quantum yield is too low for direct measurement, can be determined indirectly

1. Fluorescence derivatisation

A chemical, electrochemical or photochemical reaction converts the non-fluorescent analyte into a fluorescent compound

3 Indirect methods间接测定法（不要求）

2. Fluorescence quenching method

The analyte does not fluoresce but quenches a fluorescent compound; the decrease in intensity gives its concentration

3. Sensitised fluorescence

The analyte does not fluoresce but, once excited, transfers its energy to a suitable fluorescent reagent acting as energy acceptor, which then emits

Section 4 Quantitation

4 Multi-component mixtures多组分混合物的荧光分析法（不要求）

- For mixtures, fluorometry is more selective than UV-Vis spectrophotometry: a suitable excitation or emission wavelength alone often allows one component to be measured
- When wavelength selection alone is insufficient, use the additivity of fluorescence: measure the mixture at suitable wavelengths and solve simultaneous equations for the individual contents
- More advanced techniques — synchronous, derivative, time-resolved and phase-resolved fluorometry — together with chemometric methods allow multi-component mixtures to be determined

Section 5

New Fluorometric Techniques

不要求

Section 5 New Fluorometric Techniques

1. Laser-induced fluorometry

A highly monochromatic, intense laser is used as the source. Sensitivity is 2-10 times that of conventional fluorometry, and single-molecule detection becomes possible

2. Synchronous fluorometry

- A constant wavelength difference $\Delta\lambda$ is chosen and the excitation and emission wavelengths are scanned simultaneously, giving a synchronous spectrum
- If $\Delta\lambda$ equals or exceeds the Stokes shift, sharp narrow synchronous peaks are obtained
- The narrow bands reduce spectral overlap and stray-light interference, improving selectivity and sensitivity
- The peak height is linear in concentration, so the method is quantitative

Section 5 New Fluorometric Techniques

3. Time-resolved fluorometry

Because lifetimes differ, a delay between excitation and detection allows selective detection

4. Spatially resolved fluorometry

Spatially resolved techniques include: confocal fluorometry (pinhole effect, depth profiling); total internal reflection fluorometry (removes bulk interference, probes the interface); multiphoton fluorometry (nonlinear optics, higher resolution); near-field fluorometry (scanning-tunnelling principle, beats the diffraction limit)

Section 5 New Fluorometric Techniques

5. Single-molecule fluorometry

- Two requirements must be met: (1) only one molecule interacts with the laser in the illuminated volume; (2) the single-molecule signal exceeds the background
- Both are achieved by lowering the concentration (density) and reducing the probe volume
- Measuring quantum jumps, lifetimes and quantum yields of single molecules reports on the local environment
- The orientation of the transition dipole and the local environment reveal the structure and function of biomacromolecules

Section 5 New Fluorometric Techniques

6. Fluorescence immunoassay

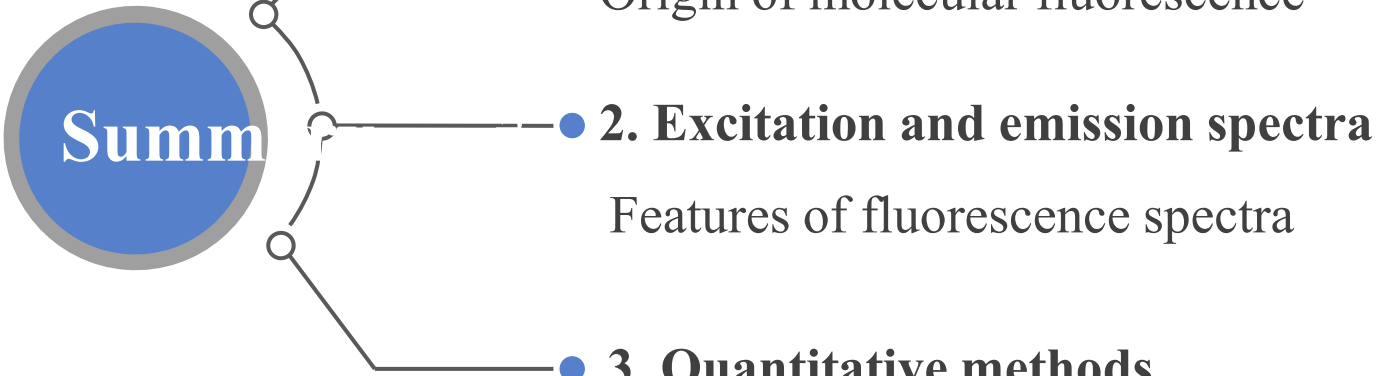
- A fluorophore that does not affect antigen-antibody activity is attached to the antibody (or antigen); after binding it shows specific fluorescence under a fluorescence microscope, confocal microscope or flow cytometer, allowing localisation, tracing and quantitation

Section 5 New Fluorometric Techniques

6. Fluorescence immunoassay

Two formats: (1) direct — labelled antibody is added to the sample and binds the antigen; simple, specific, little non-specific staining, but less sensitive and a separate antibody is needed for each antigen; (2) indirect — unlabelled primary antibody reacts first, then a fluorescently labelled secondary antibody binds it, forming a primary antibody-antigen-labelled secondary antibody complex. One labelled antibody then serves many systems

Chapter summary



本章小结

小结

1.荧光分析法的基本原理

分子荧光的发生过程

2.荧光的激发光谱和发射光谱

荧光光谱的特征

3.荧光定量分析方法

直接测定法、间接测定法、多组分混合物的荧光分析法

Thank you